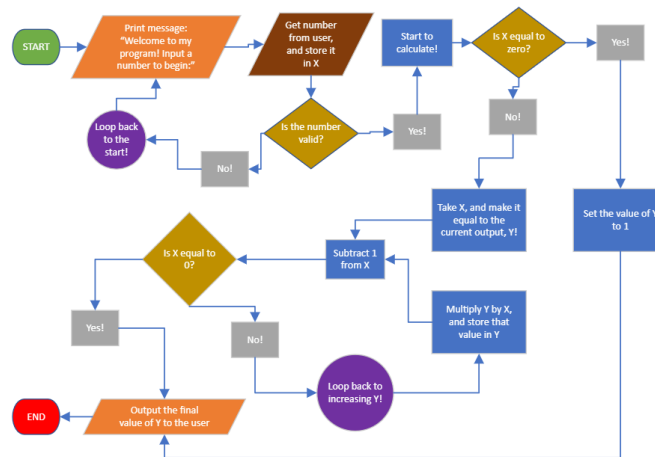


Assignment 1 – Introduction to Programming, Flowcharts



Objective

In this assignment, you will practice thinking like a programmer as you create flowcharts to break-down and solve more complex problems. Using simple logic, math, and programming structures, you will create a flowchart so that when followed under given starting conditions, a specific output is reached.

Background

In this assignment you will use simple blocks to construct a flowchart that can solve a programming problem. You have several such blocks at your disposal, including; start, end, get input, give output, process, conditionals, and loops. Using these simple blocks, you must create your own “program” (flowchart) to model a process of your choice. Refer to the document “Programming Flowchart Legend” to see the various types of blocks with their explanations. Images can be seen below:

Start and End

In your flowchart, represent the start and end of your program as follows. Start and End blocks count as 1 block each for this assignment:



Input and Output

In your flowchart, represent input and output with the blocks observed below. The words 'INPUT' and 'OUTPUT' should be replaced with a description of what is being input to or output from the program. Input and Output blocks count as 1 block each for this assignment:



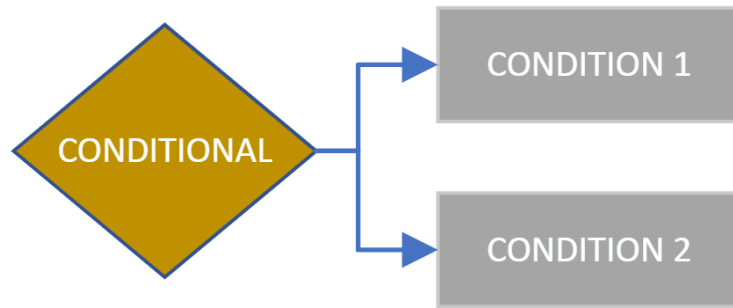
Process

In your flowchart, represent processes with the block observed below. Processes are any operations or procedures that transform or create new values. Ex. calculate the volume of a cylinder. The word 'PROCESS' should be replaced with a description of the process. Process blocks count as 1 block for this assignment:



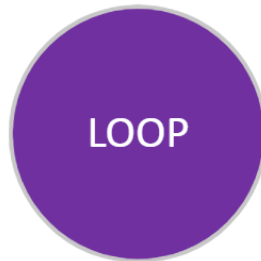
Conditional

In your flowchart, represent conditionals with the block observed below. The word 'CONDITIONAL' should be replaced with a description of the conditional (question), the text in each gray 'CONDITION' box should be replaced with a description of the condition (answer). The entire conditional block, including the conditional and its conditions, counts as only 1 block for this assignment:



Loop

In your flowchart, represent loops with the block observed below. The word 'LOOP' should be replaced with a description of how the loop is directing the program flow. Loop blocks count as 1 block for this assignment:



Instructions

Create a flow chart diagram to model the process behind a simple computer program of your choice. When choosing your program to model, pick a process that you understand on a fundamental level, and apply programming concepts to it. If you pick a program that is too difficult, you will have a hard time constructing it out of basic programming blocks. It is highly recommended that you use Microsoft Visio software to illustrate your flowchart, but you are free to use any software so long as you follow the shapes and colours of the example diagrams above.

Using the above blocks, create a flowchart that breaks down your selected process. In order to get full marks, you are expected to use at least 14 blocks, counting blocks as defined above. Every block must contribute to your program in a way that removing it would take away from clarity. Unnecessary blocks may be subtracted from your total.

The rubric on onQ contains a more specific breakdown of the marking.

Here are some simple programs as possible ideas:

- Math operations of simple to moderate complexity
 - Locating roots of a parabola
 - Calculating the volume and surface area of a box with dimensions x , y , and z
- Economics problems
 - Calculating return on investment
 - Calculating the price of an object with inflation
- A simple program with logic you can understand
 - Vending machine item selector
 - Traffic light system

Explain in Plain English

The last thing you should make for this assignment is a short explanation of your flowchart in plain English. **Write it once your flowchart is finished**, not before.

Explain what the program in your flowchart does. **Describe what the program accomplishes and how its parts work together to do it**, not what each block does one at a time. A block by block translation of your flowchart into English is not what is being asked for here. Someone who has never seen your flowchart should be able to read this and understand what your program is for. A short paragraph is enough.

Point form is fine, and you are being graded on the content, not on your writing skills. Put your explanation in a text box in your flowchart. Use the heading 'Explain in Plain English'. Depending on space, you may want to put it below your flowchart such that it is on a separate page when you export it as a PDF.

When you are done, ask yourself whether someone who has never seen your flowchart could read your paragraph and tell you what your program is for. If they could, your explanation has done its job.

Submission Instructions

Save your flowchart (containing the plain english explanation) as a pdf and submit it to the assignment page on the course onQ before the deadline posted there. Make sure the file follows the naming convention of:

`APSC141assign1_lastName_firstName_studentNumber.pdf`

For example, Jane Smith with student number 20481234 would submit:

`APSC141assign1_Jane_Smith_20481234.zip`

Submissions may receive a 1 mark penalty for containing any incorrectly named or created files.

Late Policy

Late submissions are penalised **10% of the assignment mark per day**, up to a maximum of three days. **After three days, the assignment receives a mark of 0.**

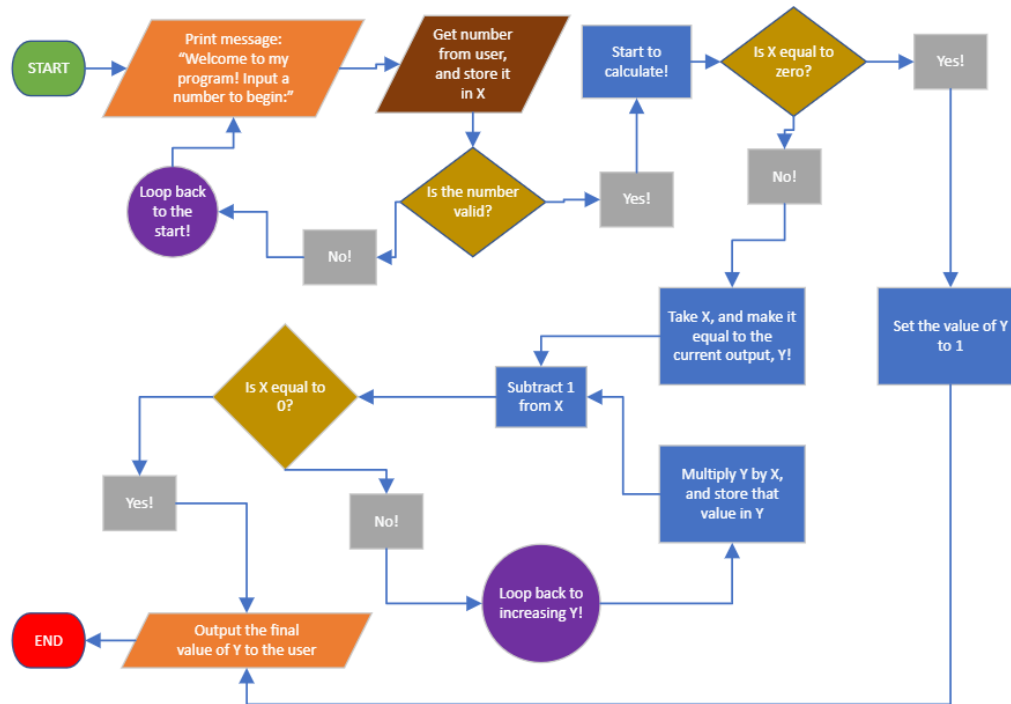
Use of Generative Artificial Intelligence (GenAI) Tools

Queen's GenAI tool, QUAN, is permitted in this course.

Other GenAI tools are NOT permitted in this course. Use of GenAI tools other than QUAN is not allowed in any part of student work for this course. Submitting AI-generated content constitutes a Departure from Academic Integrity as defined by University Academic Integrity procedures.

Example

Try to understand the program represented in the following flowchart:



Which is explained in plain english below:

This program works out the factorial of a number. It asks for a number and checks that what was typed can be used, prompting the user to enter a new value if it can't be. Zero is a special case dealt with on its own, because the factorial of zero is one, and doesn't match the pattern used for every other number. For those numbers, the program multiplies the inputted value by each smaller whole number one at a time, keeping a running total. This process counts down until it reaches zero, at which point the running total is the factorial, and it is shown to the user.